

EMPOWERING THE DEVELOPER

How to Build an E-Commerce Application using *J2EE™*

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Code Camp Engineer



Code Camp Agenda

- J2EE™ & Blueprints
 - Application Architecture and J2EE
 - Blueprints E-Commerce Application Design
- Enterprise JavaBeans™ (EJB™)
 - Designing and implementing the Business Model:
 - Entity Beans, Stateless and Stateful Session Beans
- Servlets , Java ServerPages™ (JSP™)
 - Designing and architecting the "View" or Presentation logic
- Assembling and Deploying the J2EE™ Application:
 - Transactions JTS™ and JTA™
 - Describe how the sample application uses the transaction capabilities of the J2EE™ platform to simplify component development.
 - Security How to use J2EE™ security to safeguard your ecommerce application

Download Code Camps

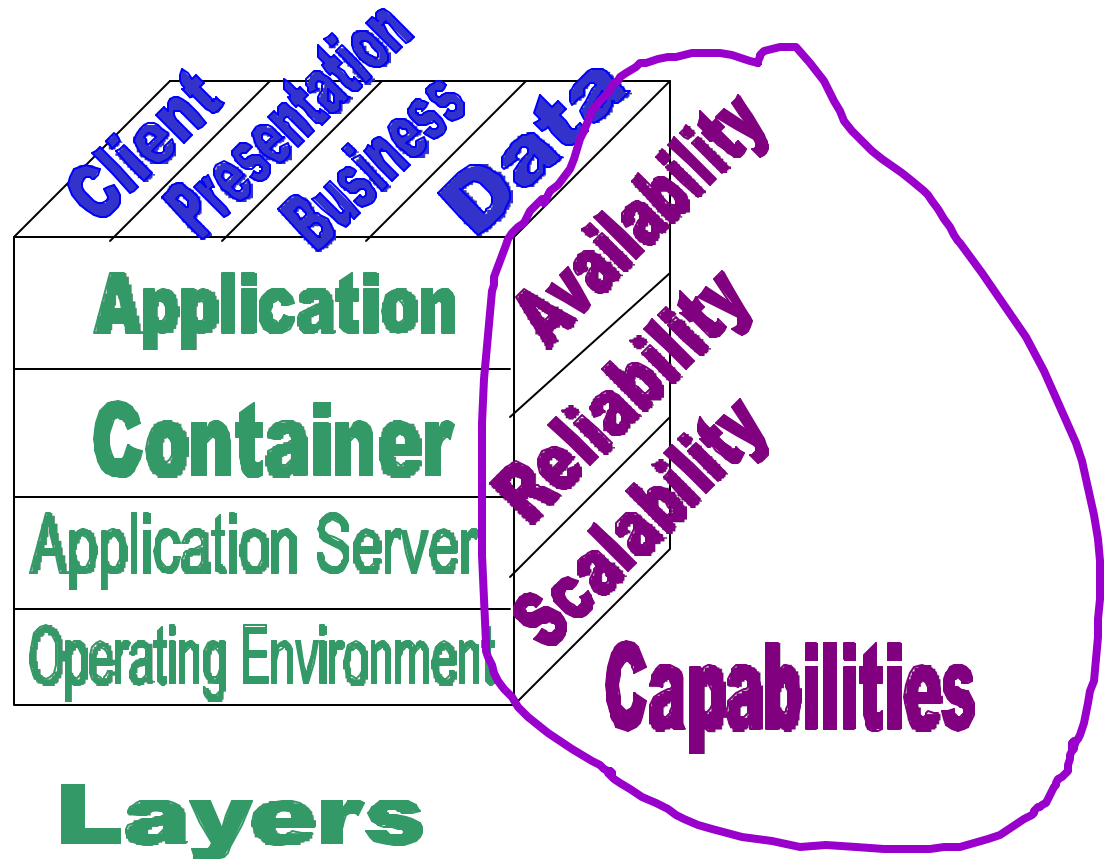
- All slides with notes are on your CD
 - You can **download** this and other Code Camps :
 - EJB, Java Performance, J2ME, JINI, JMS, JSP, XML
- <http://www.sun.com/developer/evangcentral>
- under Events: Developer Code Camps
 - ALSO
 - You can also ask questions there
 - Listen to audiocasts

J2EE™ & Blueprints Agenda

- *Application Architecture and J2EE*
- Sun BluePrints™ Design Guidelines
- Architecture of the sample application

Architecture and the Cube

Tiers



Architecting your application Using **Tiers** and **Layers** affect the application **Capabilities**

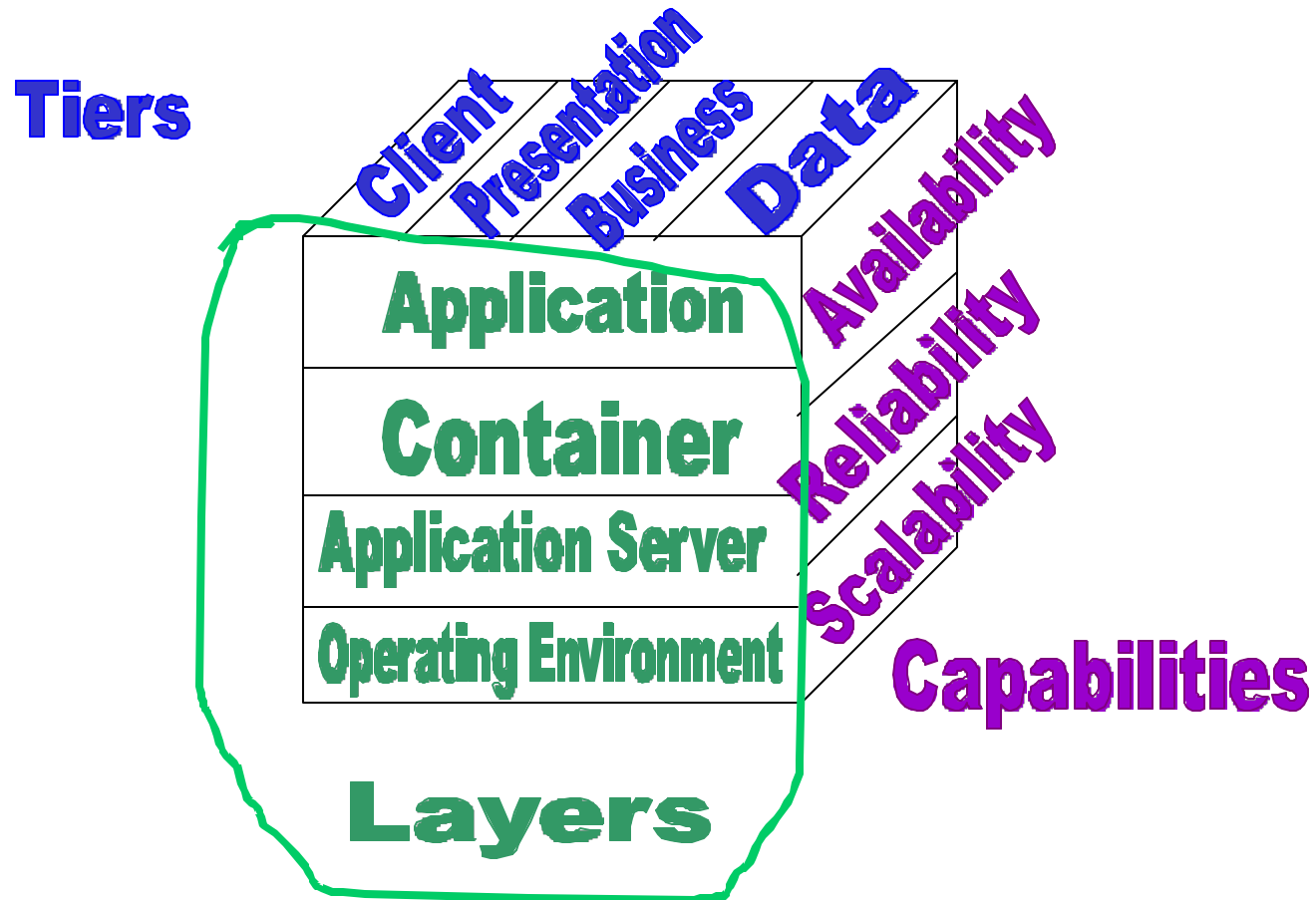
Capabilities = Non-Functional System Requirements

- **Availability**: service is accessible
- **Reliability**: consistency of application and transactions
- **Capacity**: can serve # users
- **Extensibility**: can add functionality
- **Flexibility**: can support changes
- **Performance**: good response time
- **Scalability**: can support increased load

Example Architecture Requirements

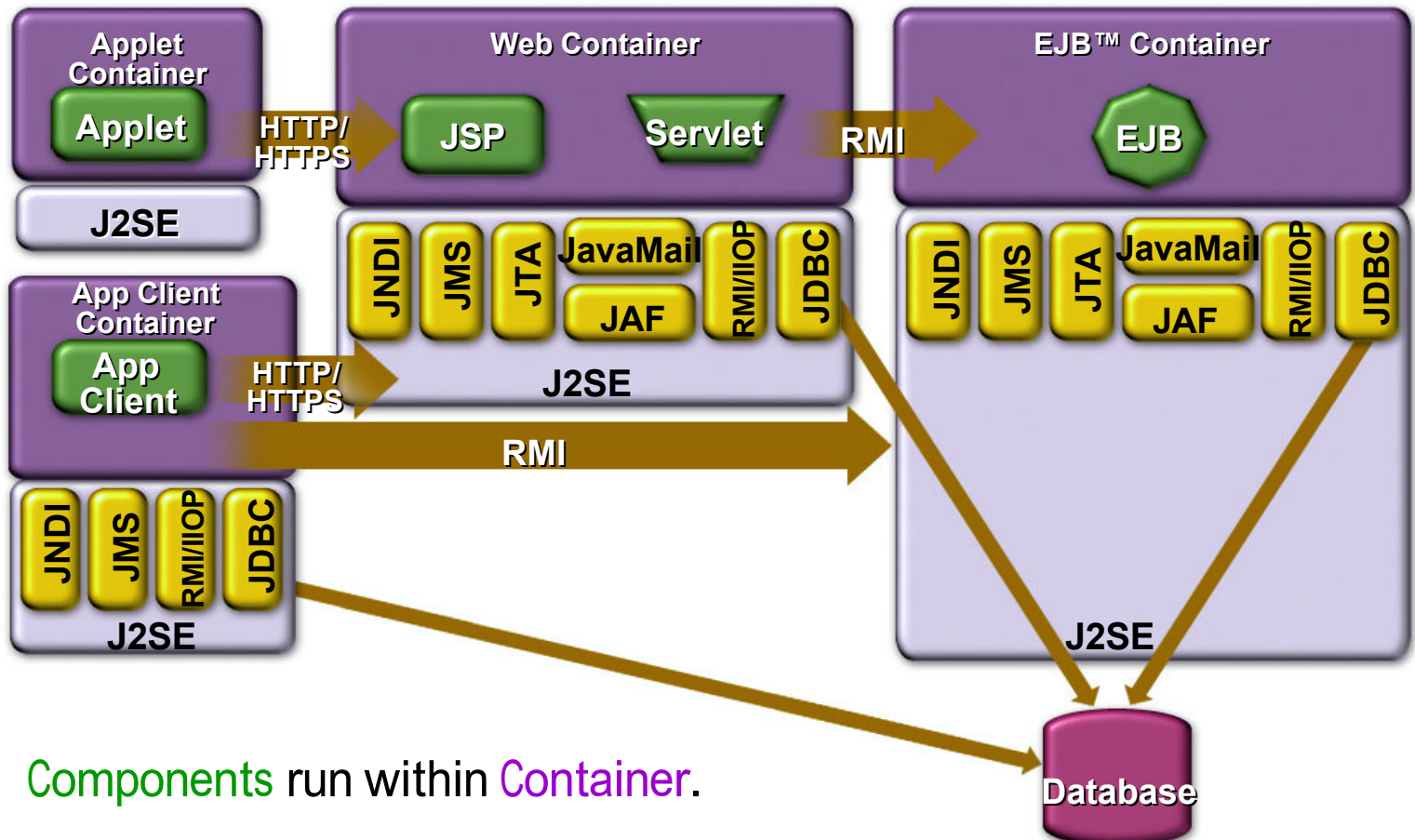
- E-Commerce Apps:
 - Many Clients,
 - lots of **reading**,
 - **less updating**,
 - low contention
 - **Need High performance and scalability** for Reading data
- Insurance Intranet App:
 - Fewer Clients, **Low concurrency**
 - High transaction isolation level, **Need Higher Consistency**
 - **Reliability** of Updates to DB more important than performance for Reading data

The Relationships between the Elements of Architecture



Architecting your application Using Tiers and Layers affect the application Capabilities

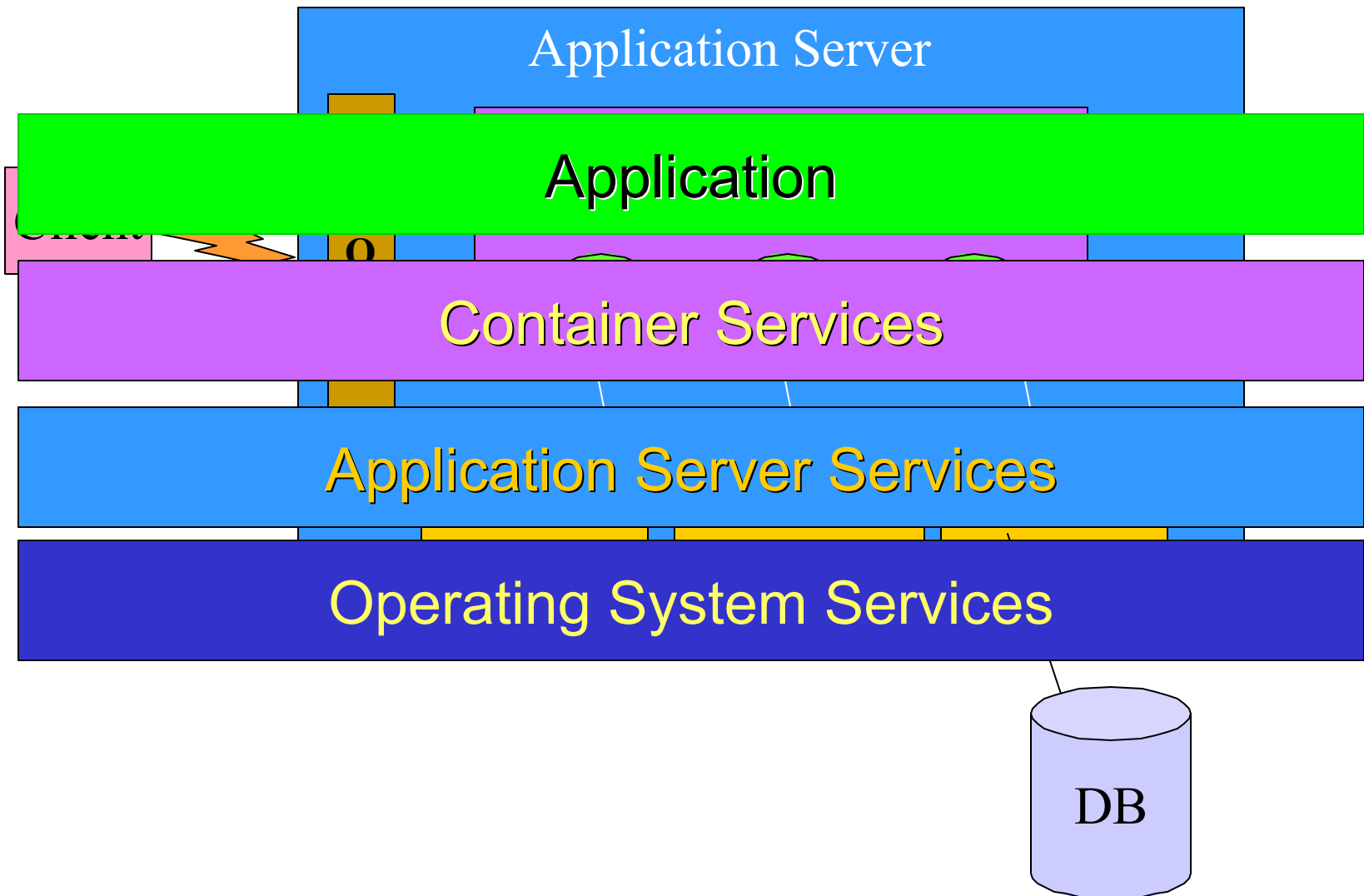
J2EE™ Platform Specification



Components run within Container.

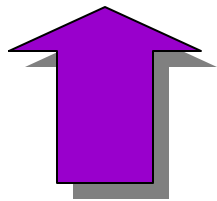
Container provides Runtime environment, J2SE™ & J2EE™ APIs, and remote communication

Layers of Services: The Application Server provides services to the Container, which in turn provides services to the EJB.

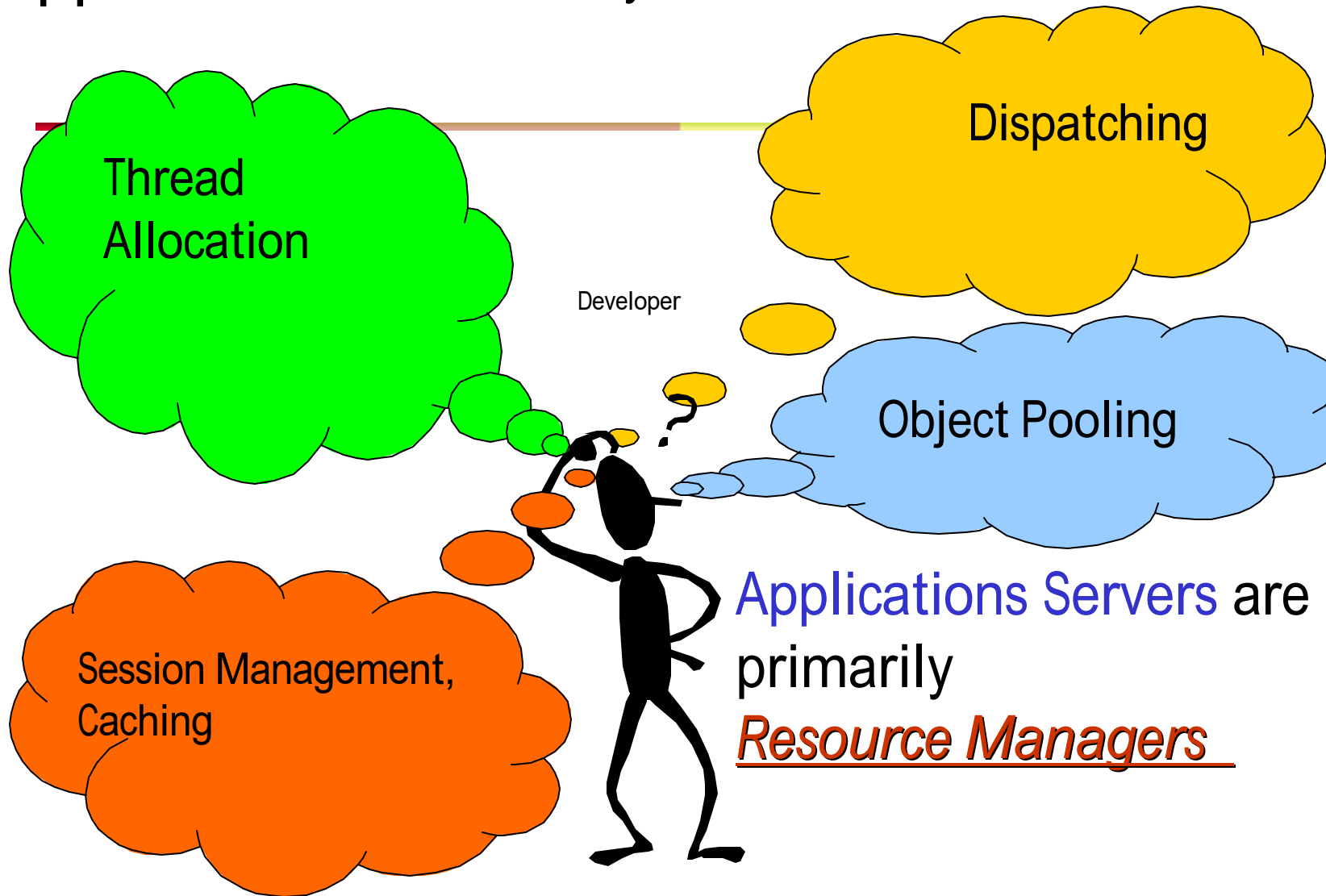


Layers

- Layers are **abstractions** of underlying implementation
- Each Layer **hides implementation details** of layer below
- The **Container** provides components **services** of the layers below
- **Advantages:**
 - Easier development, maintenance for high performance, scalability
 - Does **not** require **developer** to **know** complex details of **distributed protocols, concurrency, transactions, load balancing, security, resource pooling and sharing...**



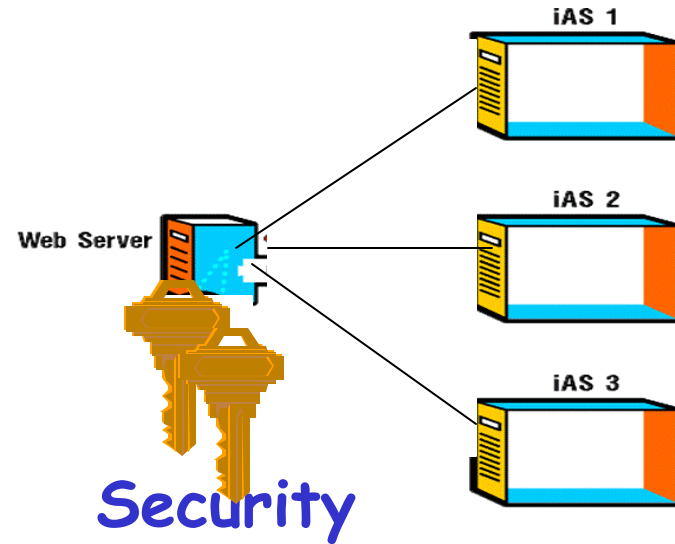
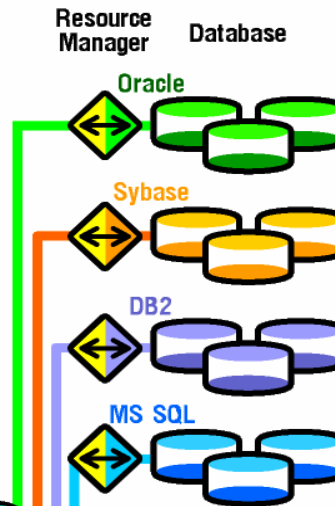
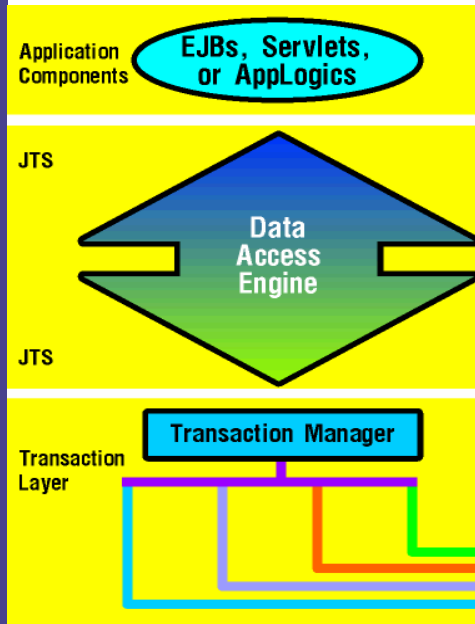
Application Server : Key Services



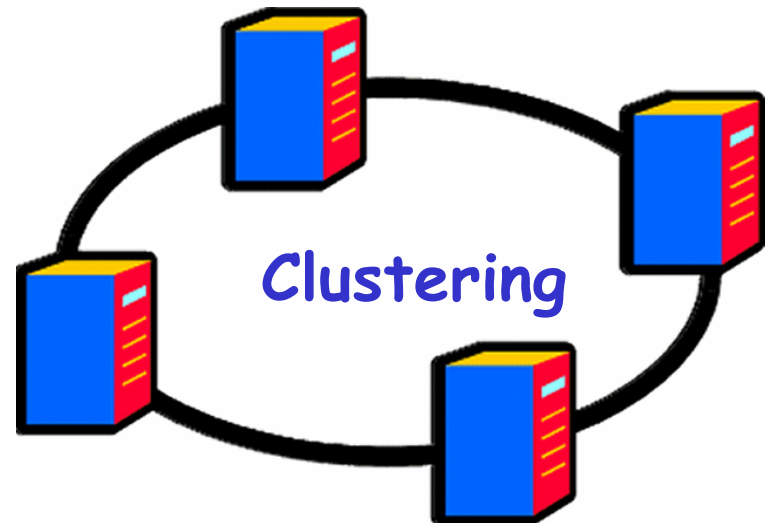
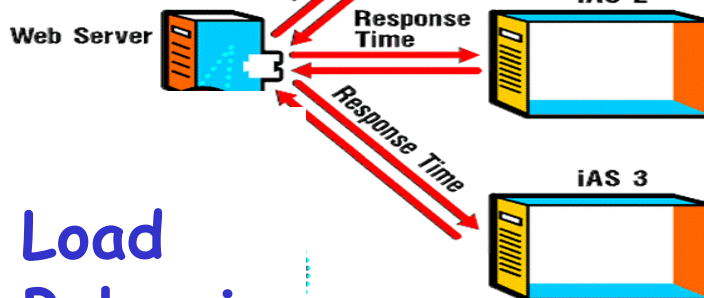
Applications Servers are primarily *Resource Managers*

Applications Servers Manage: dispatching, thread allocation, pools, caches, load balancing, clusters

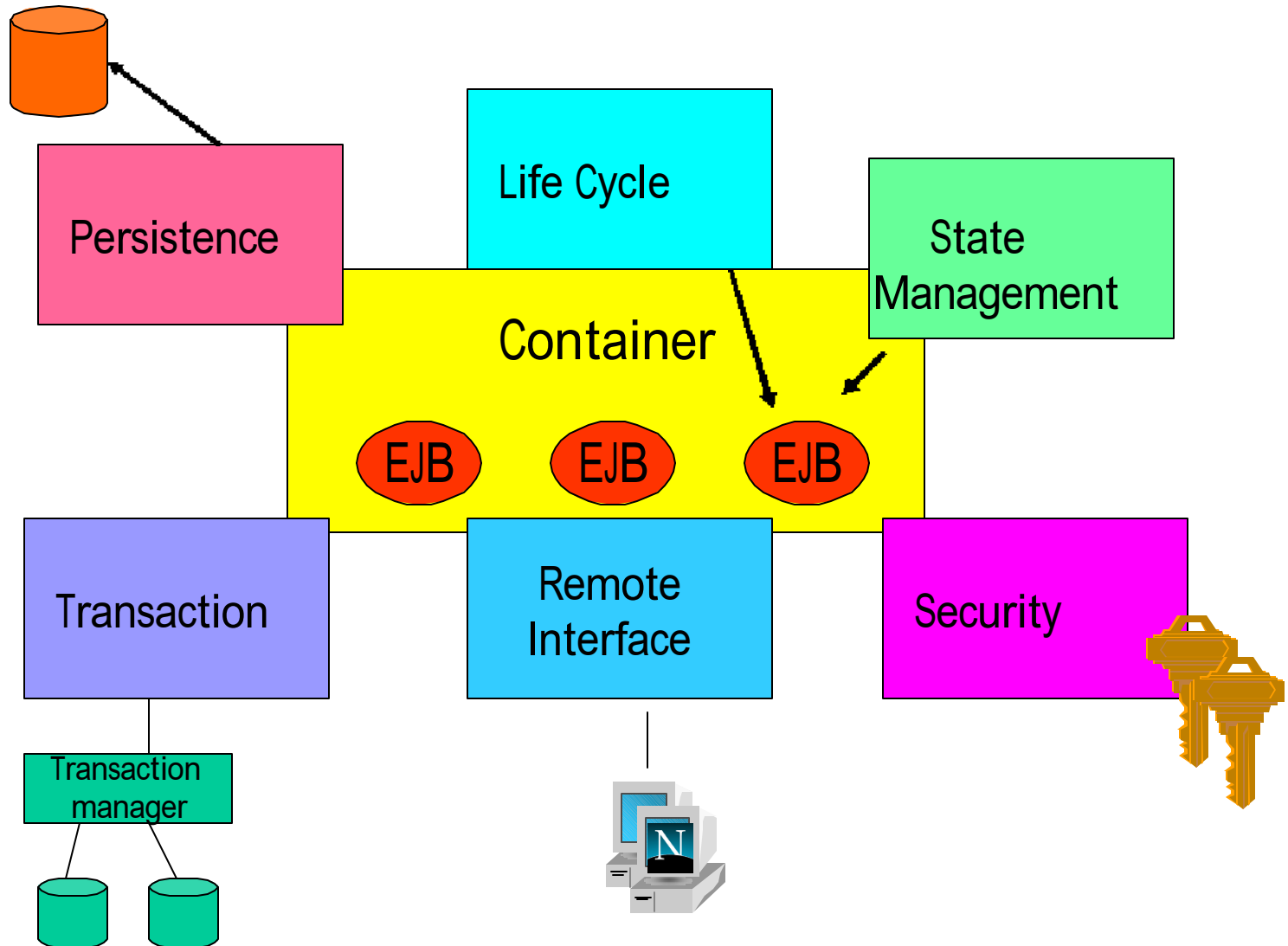
What features might you look for in an Application Server?



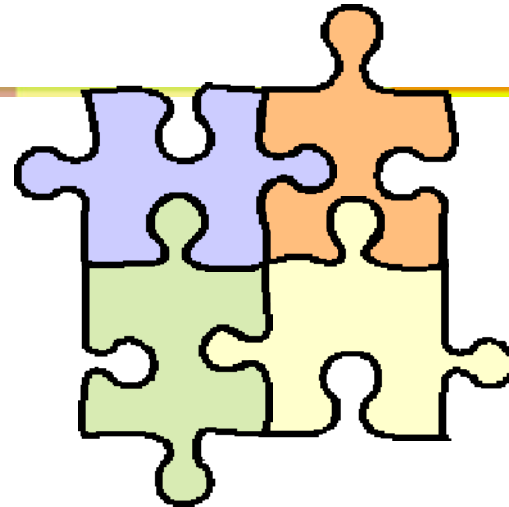
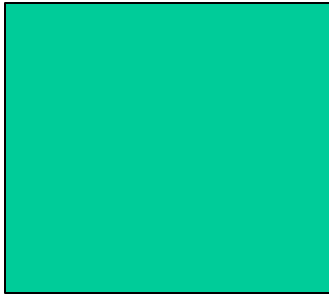
Transaction Management



EJB Container Services



Components



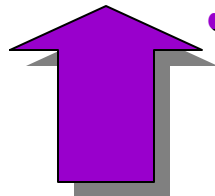
Monolithic:

1 binary file

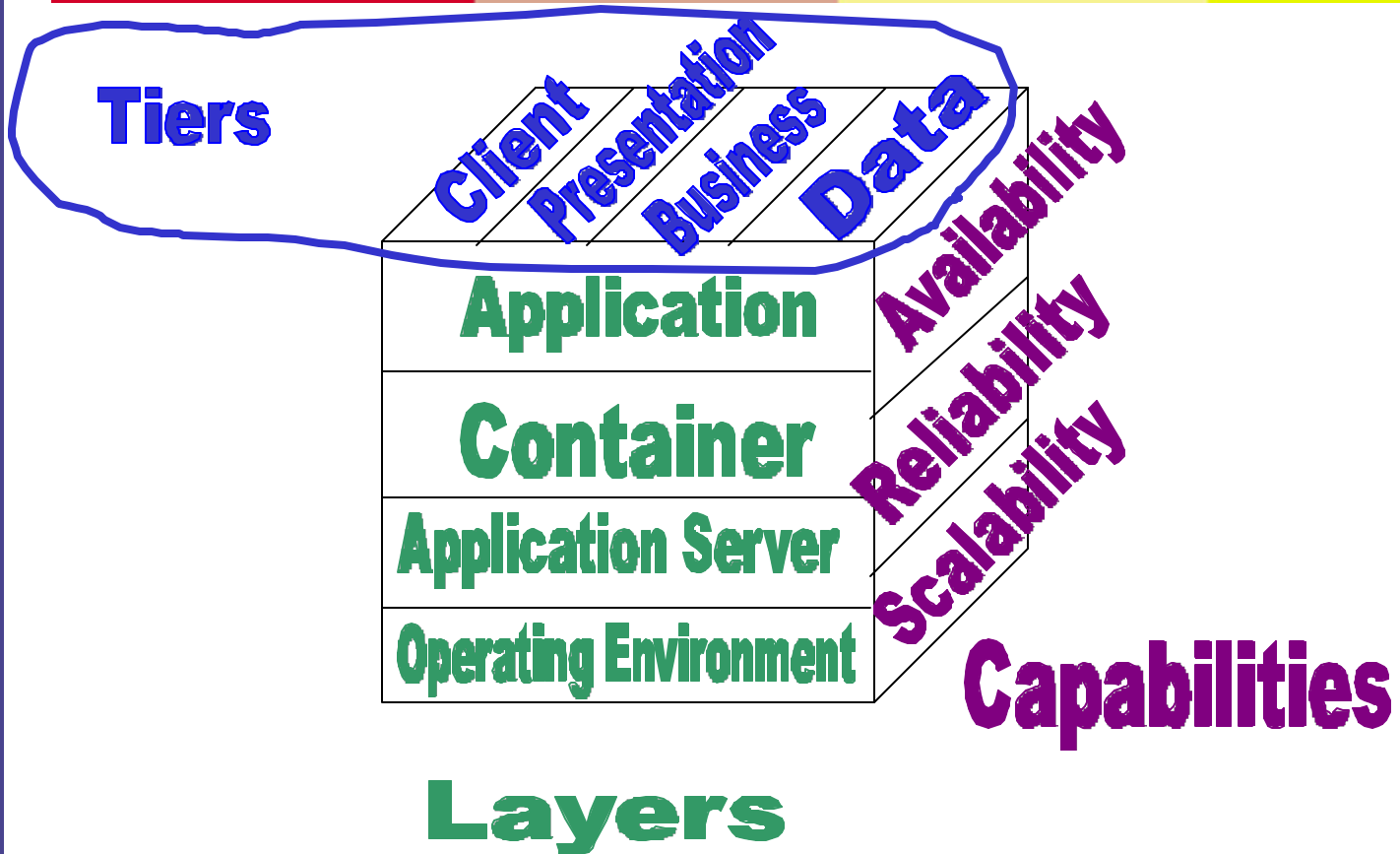
Recompiled &
linked for any
changes

Components:

- plug-gable Parts.
- Implementation separated from **Interface**, only Interface published
- hides implementation details
- Better **design**, easier **updates**
- Better Flexibility, Extensibility



Architecture and the Cube

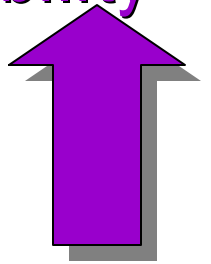


n-tier Architecture with J2EE™



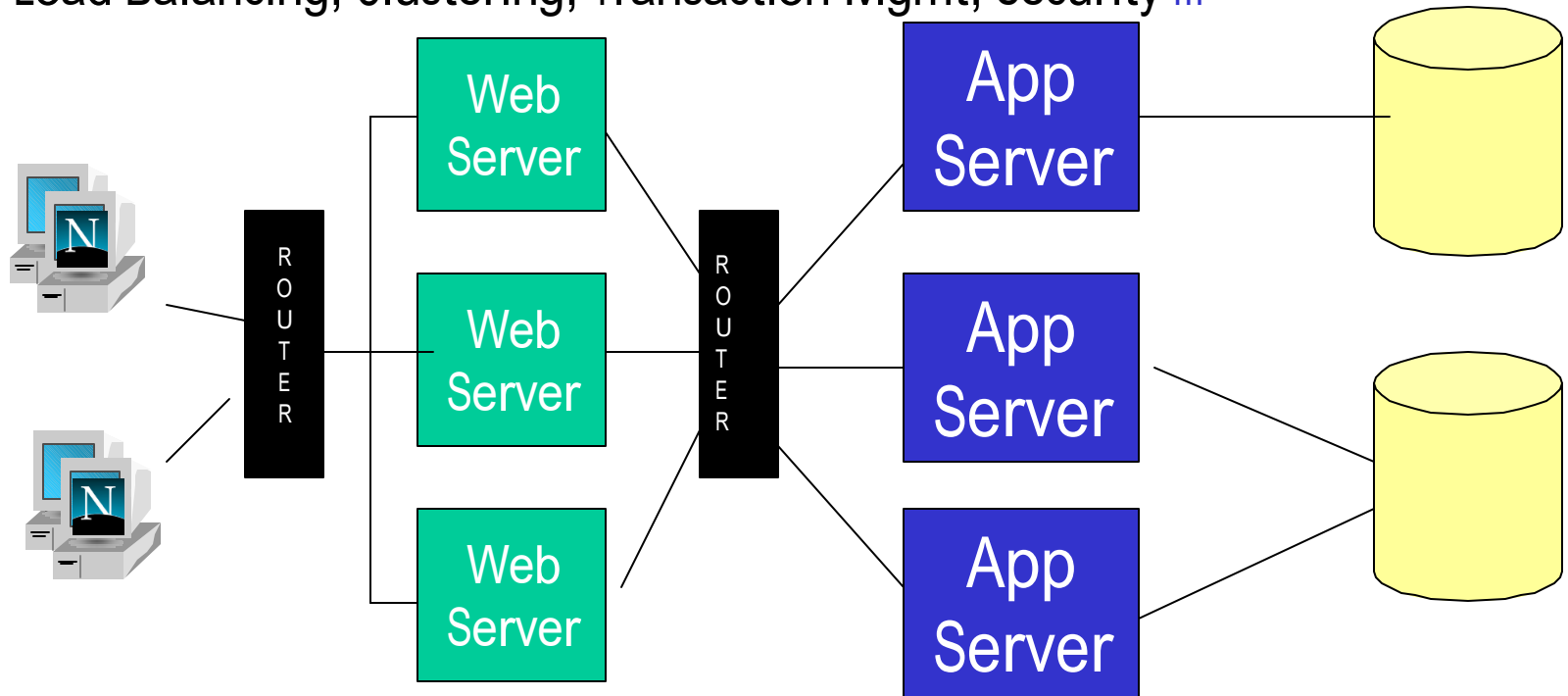
Tiers

- Tiers separate functionality
 - Presentation Logic, Business Logic, Data Schema
- Advantages:
 - One tier can be changed without changing the rest= easier updates
 - Lower deployment and maintenance costs.
 - Capabilities go up: Flexibility , Extensibility

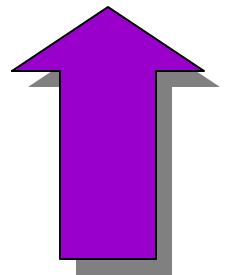


Capabilities Non-Functional Requirements

With Good Architecture, Layering, Tiers,
Load Balancing, Clustering, Transaction Mgmt, Security ...



Reliability, Availability, Extensibility,
Flexibility, Scalability, Goes up



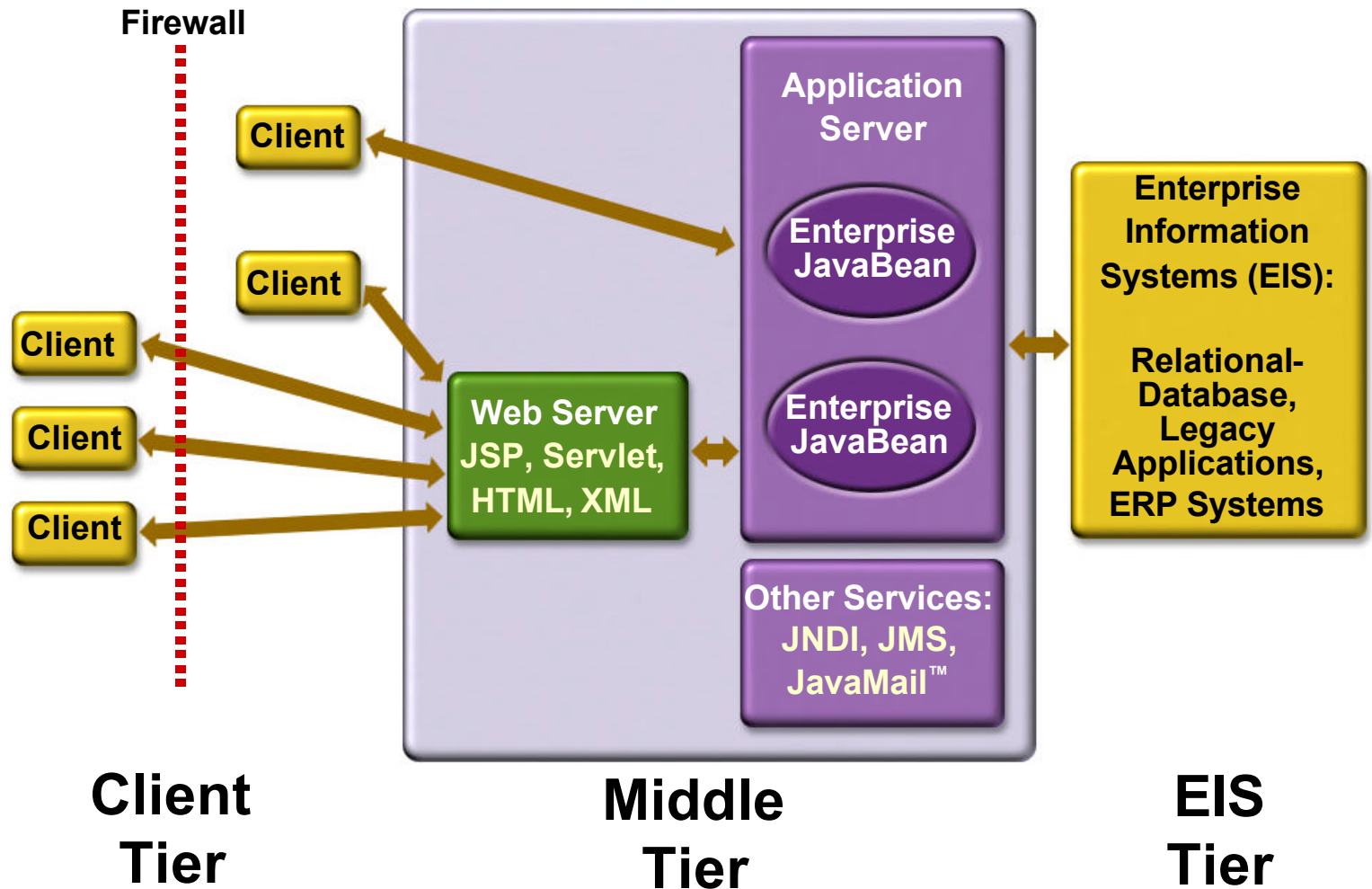
Why J2EE™?

- Gives you the **Advantages** of a **Standardized Layered, Tier** Architecture:
- **Simplifies** the complexity of a **building n-tier applications**.
- **Standardizes** an **API** between **components** and **application server container**.
- J2EE™ **Application Server and Containers** provide the **framework Services**

Agenda J2EE™ & Blueprints

- J2EE Architecture
- *Sun BluePrints™ Design Guidelines*
- *Architecture of the sample application*

The J2EE™ Platform Has a Rich Set of APIs

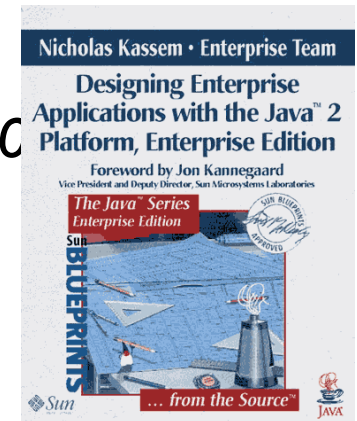


Why J2EE™ Blueprints

- Rich APIs → More than one way of doing things
- Similar Problems → Need to leverage time-tested solutions
Don't reinvent the wheel
- Blue Prints → Provides Answers, Examples of Best Practices

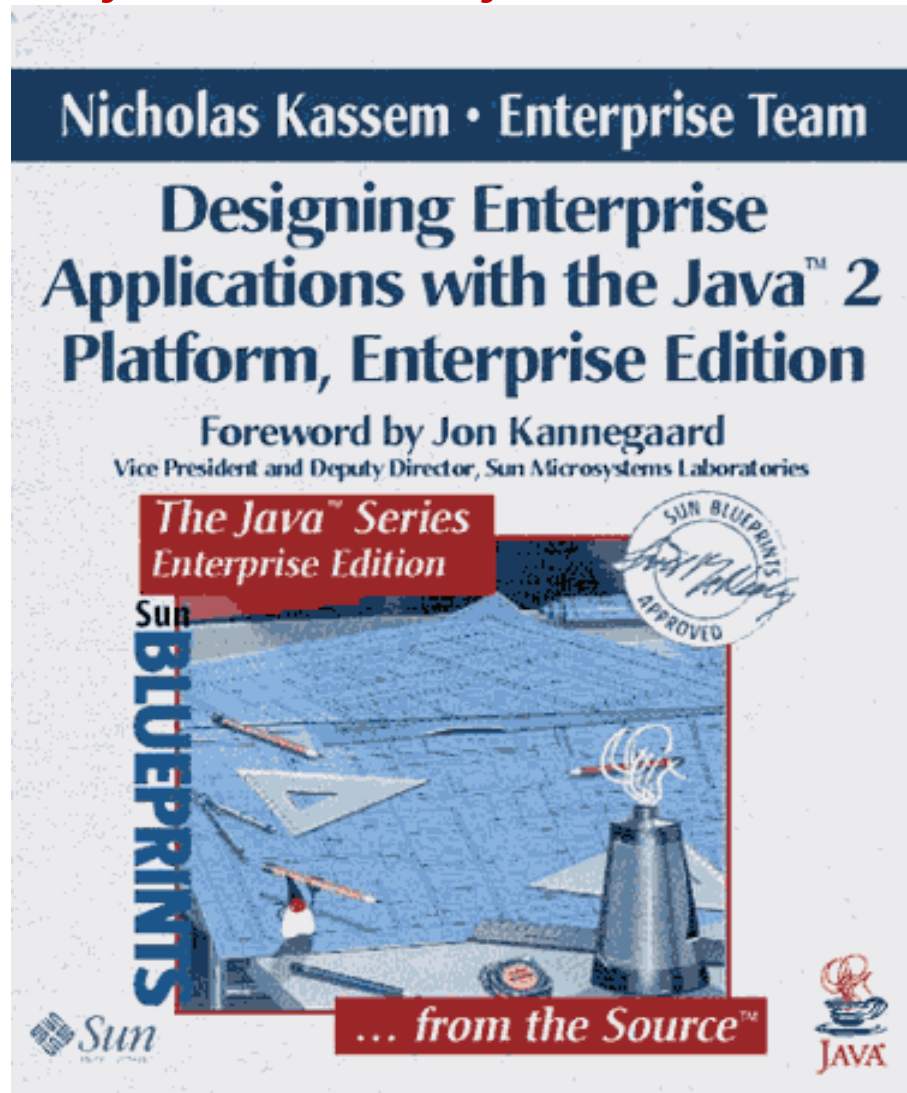
J2EE™ Blueprints

- Guidelines and Best practices for designing *n-tier* enterprise applications with J2EE™
 - J2EE™ blueprints include:
 - Book, *Designing Enterprise Applications with the Java™ 2 Platform, Enterprise Edition*
 - Application (source code)–
 - “Java Pet Store”
- FREE:
- <http://java.sun.com/j2ee>

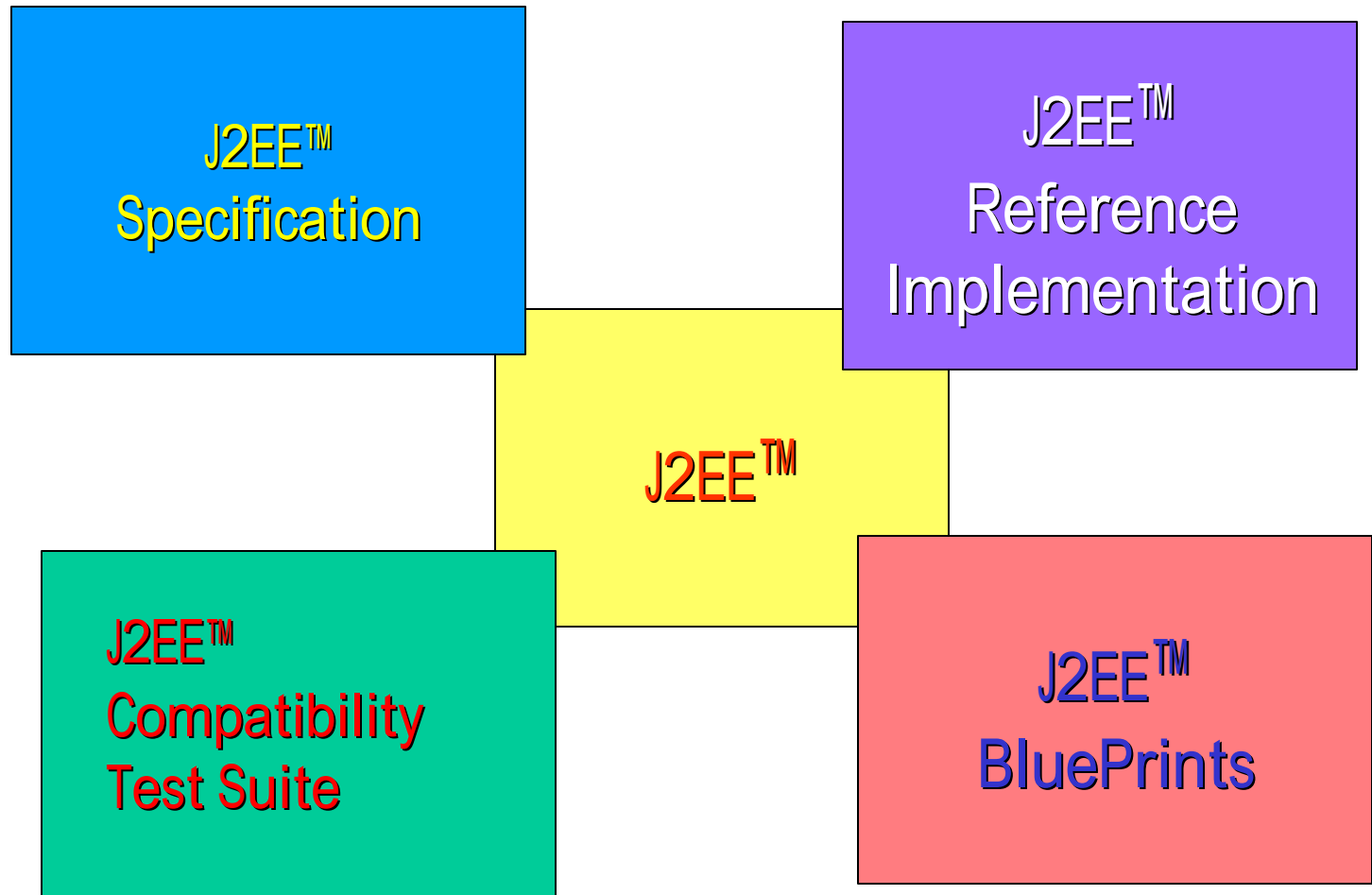


Blueprints

FREE: <http://java.sun.com/j2ee>



J2EE™ Deliverables



"Java™ Pet Store" Demo



Some Key BluePrints™ Questions -Issues Faced by Developers

- When to use Servlets versus JavaServer Pages™?
- Where to store client session state: client-tier, web-tier or EJB™ tier?
- Access Data Base Resources via EJB™ or directly from JSP™ JavaBean ?
- When to use EJB™ components?
- When to use Session Beans and when to use Entity Beans?

Design of Sample Application

Blueprints Choices:

Blueprints Choices:

Client tier

- **HTML** Clients for End-User Web Access
- **XML** for Data externalization (Order procurement)

Web Server tier

- JavaServer Pages™ (**JSP**)
for dynamic content generation

Application Server tier

- **EJB**™ for business logic

J2EE™ Blueprints Guidelines

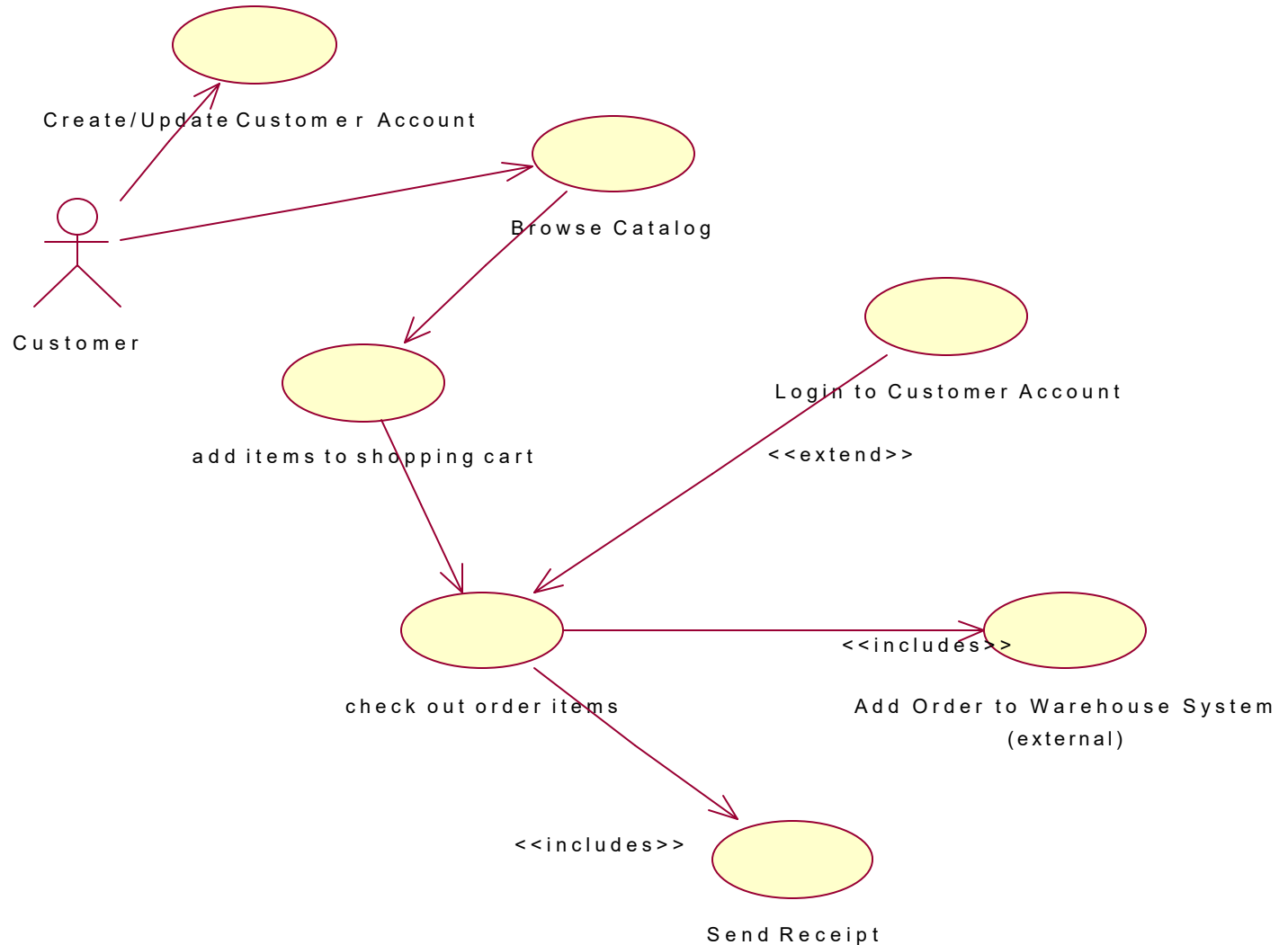
Process for developing an application from requirements, to design, to implementation

1. Scenarios or Use Cases
2. partitioning of functionality into modules
3. Application architecture using Model-View-Controller:
 1. assignment of functionality to tiers:
 1. Design of the Model
 2. Design of the View
 3. Design of the Controller
 2. object decomposition within tiers

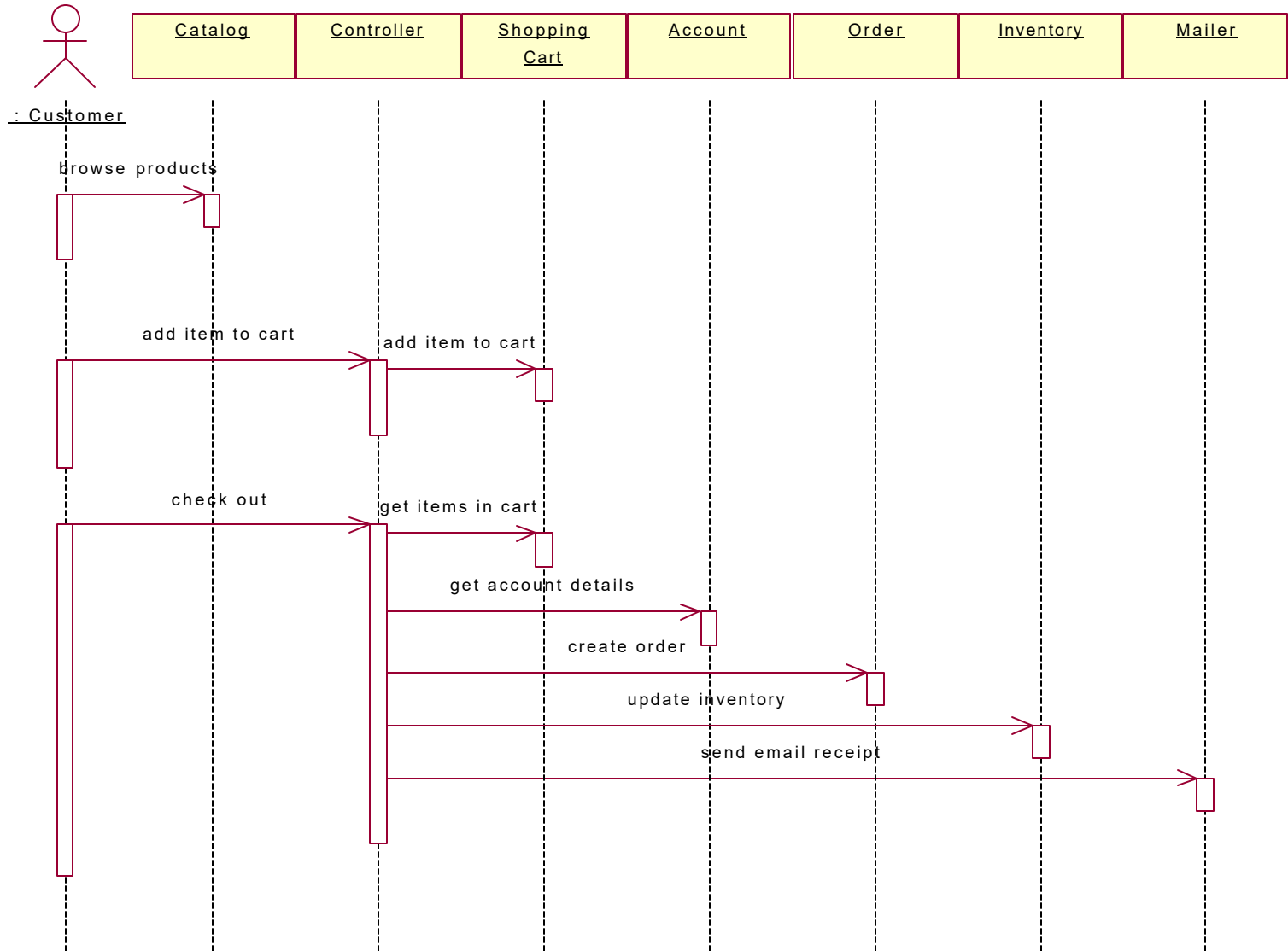
1) Sample Application – Shopping Scenario

-
1. Web-based browsing of a product catalog
 2. Creation and maintenance of a shopping cart
 3. User account creation/ Login
 4. Placing orders
 5. Secure order processing
 1. B2B transactions
 2. Externalisation of order data (expressed in XML)
 3. Order confirmation using e-mail

1) Use Case Shopping Scenario



Shopping Scenario Sequence Diagram

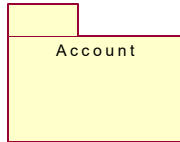


J2EE™ Blueprints

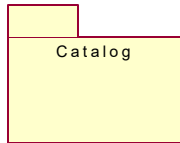
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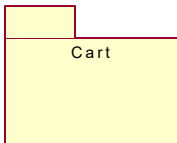
2) Divide the Application Into Modules Based on **Services (Functionality)**



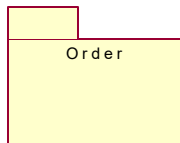
User Account Module: tracks user account information



Product Catalog Module: provides search for products, and product details

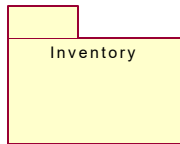


Shopping Cart Module: allows user to save selected items for the session.

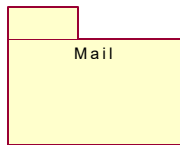


Order Processing Module: performs order Processing when user buys the items in the cart.

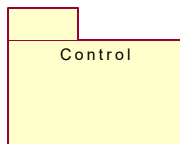
2) Divide the Application Into Modules Based on **Services**



Inventory Module: maintains information on the number of each type of product in stock.

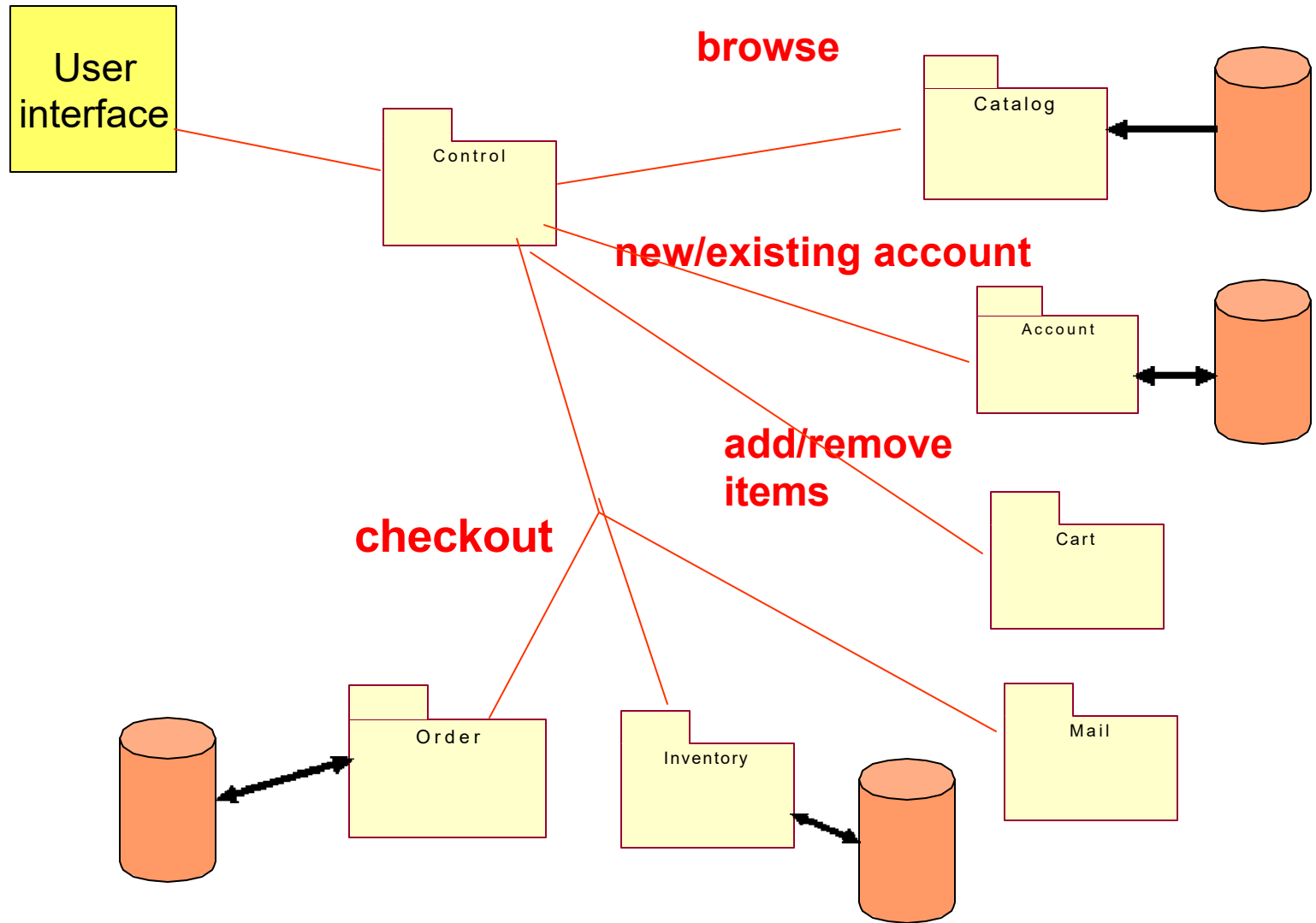


Messaging Module: sends confirmation receipt messages.



Control Module: control interactions between user (browse, add items, check out) and business objects.

Interrelationship of the modules



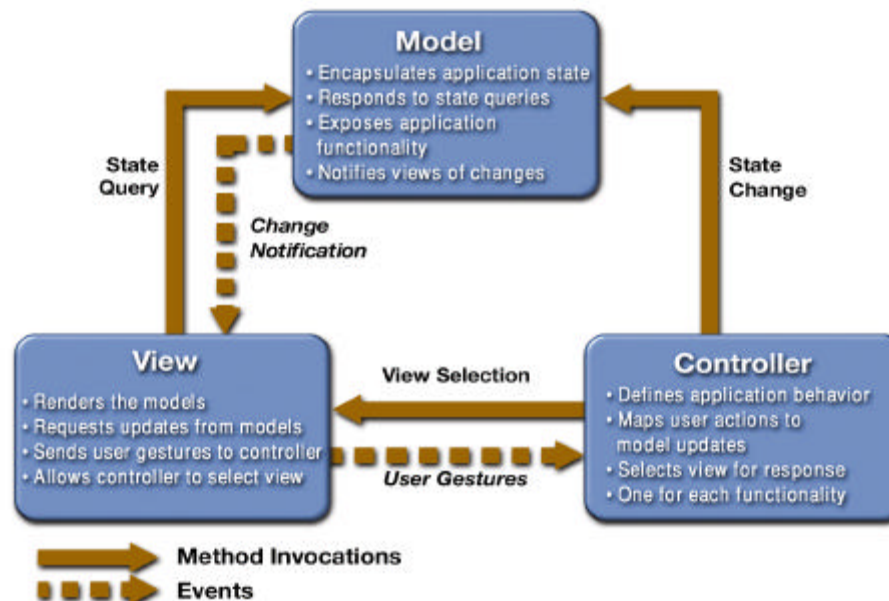
J2EE™ Blueprints

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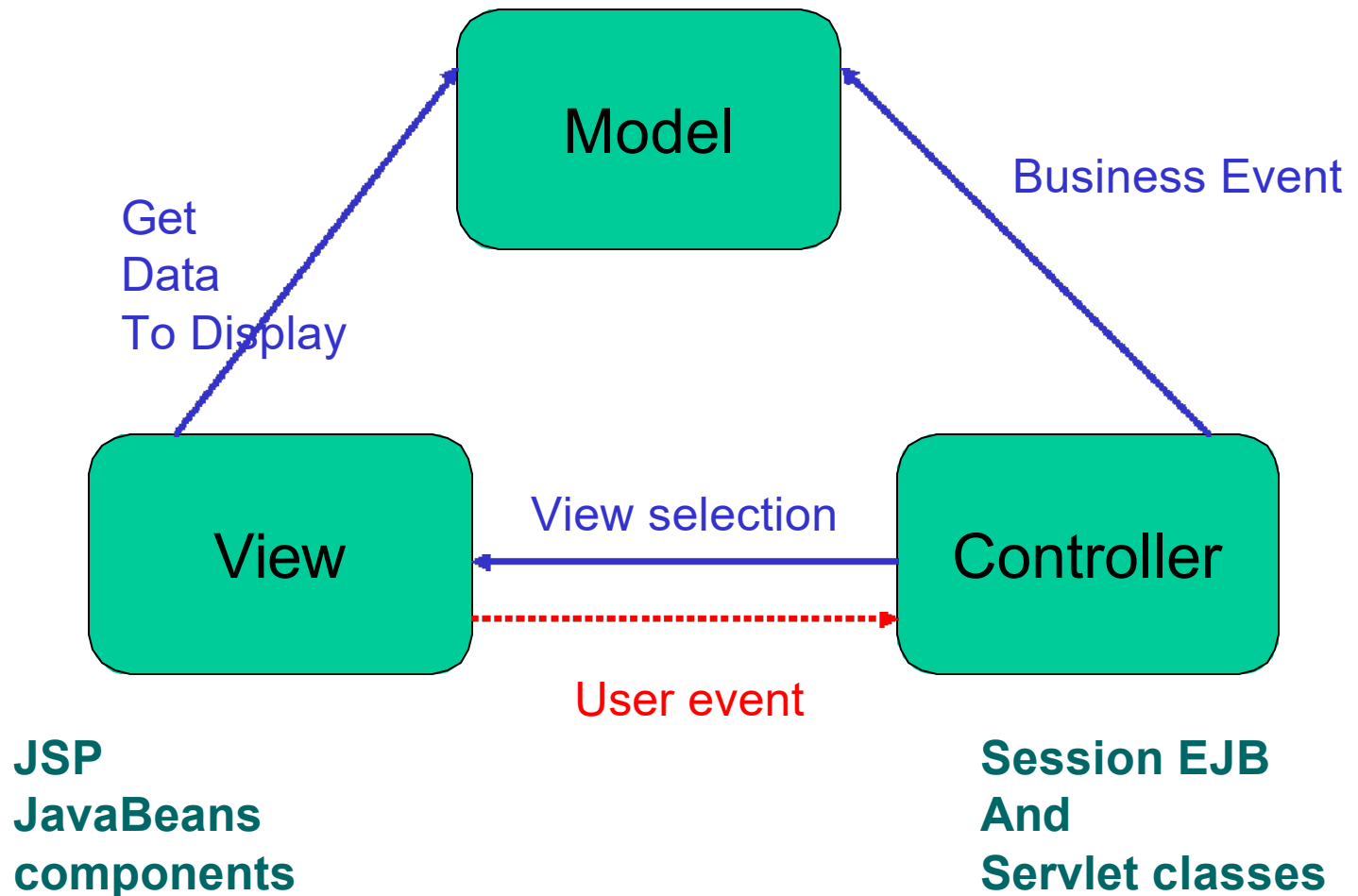
Divide Application Objects into 3 categories:

- Model View Controller (MVC) UI Pattern
 - **Model** represents the application **business data and rules**
 - **View** is a screen representation of model, **presents the user interface**
 - **Controller** **mediates** their **interactions**

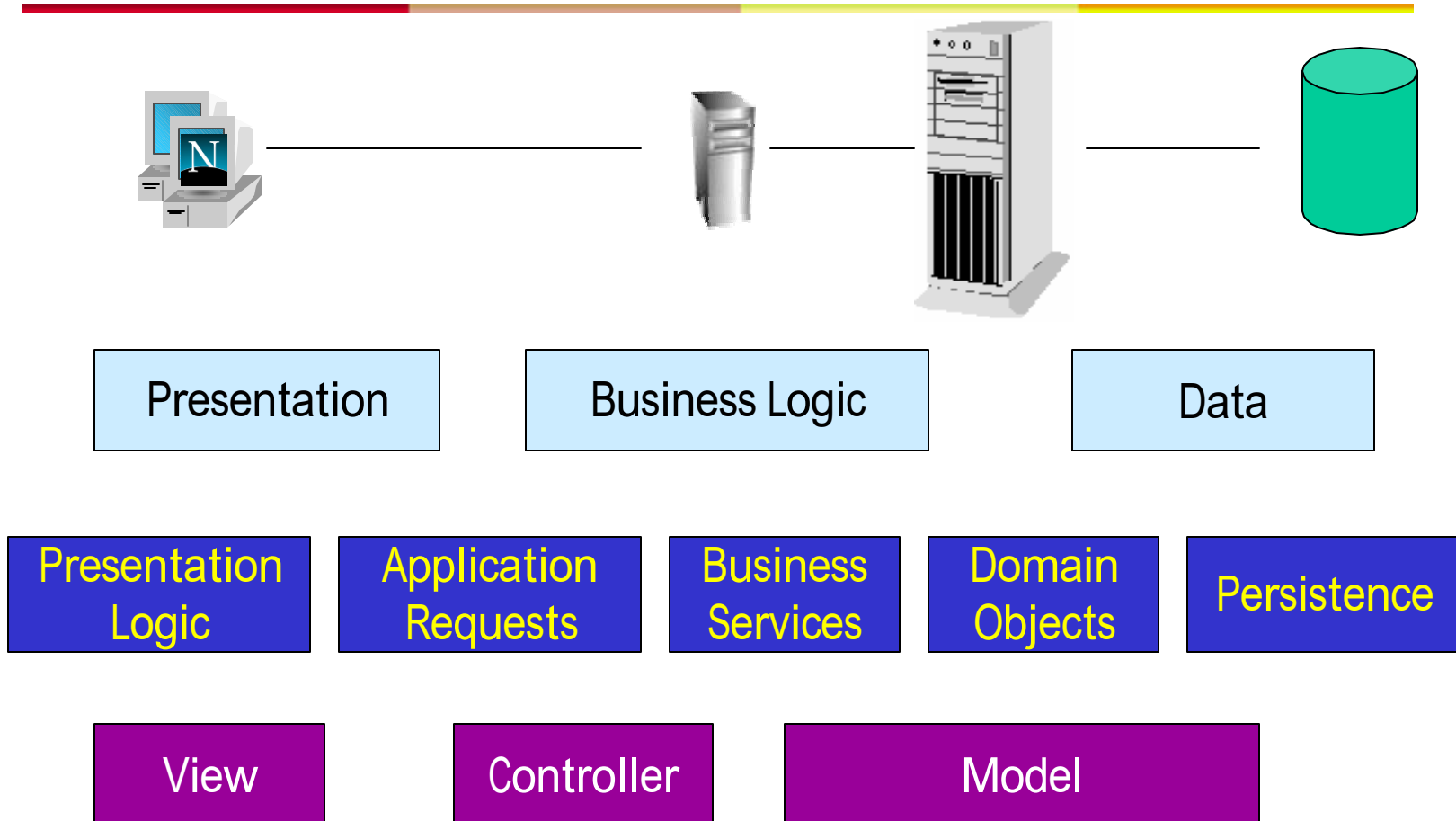


Model View Controller Design Pattern

Entity EJB and Session EJB



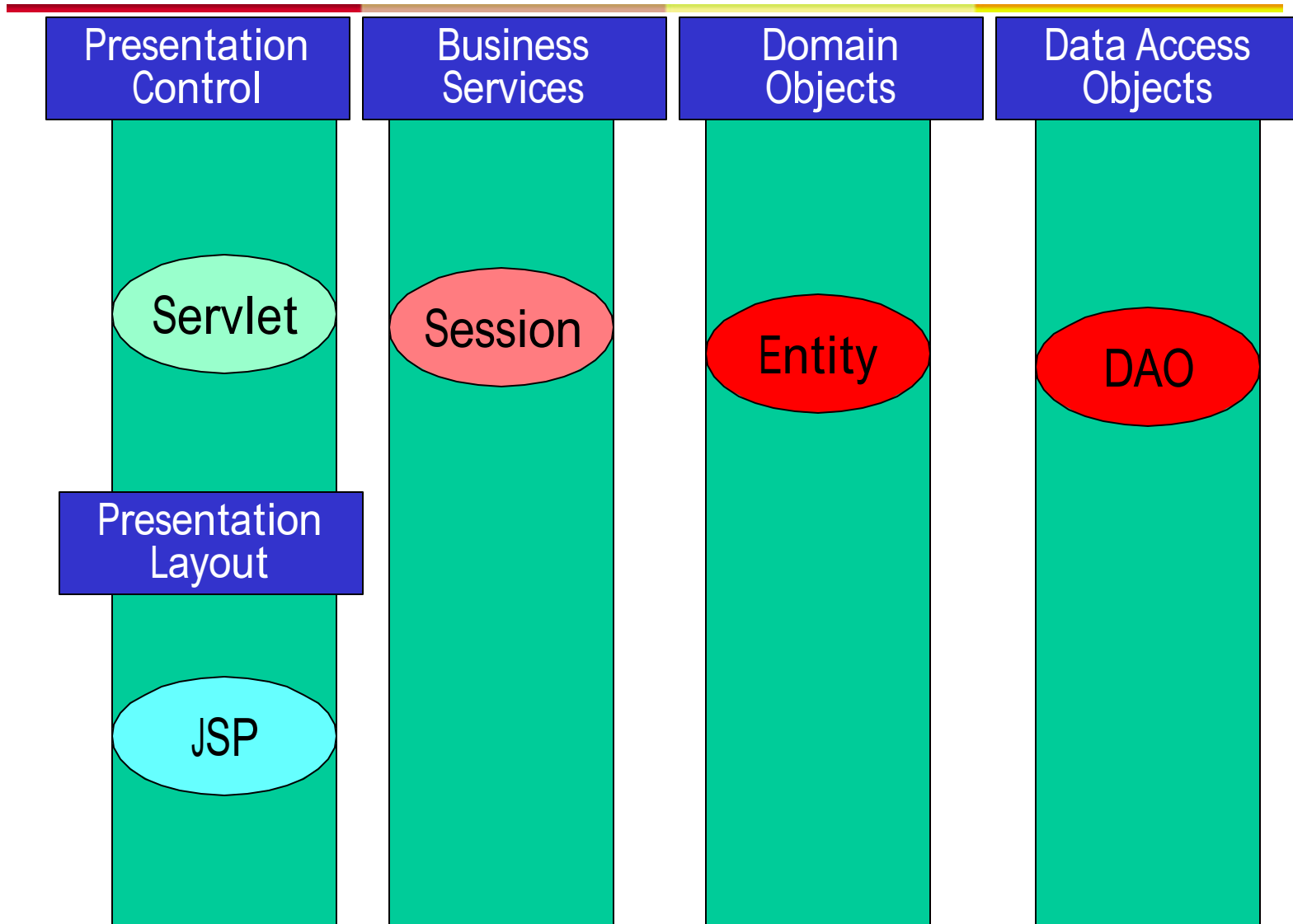
Assignment of Functionality to Tiers



MVC With J2EE™ Technology

- ï Build **View** in the **Web**-tier
 - ï Use **JSP**™ components for **presentation layout**
 - ï Custom tags and **JavaBeans**™ components for **presentation logic**
- ï Build **Controller** in the **Web / EJB**™ tier
 - ï **Servlet** for **dispatching**
 - ï **Session** bean for **control within business objects**
- ï Build **Model** in the **EJB**™ tier
 - ï **Session** beans for **Business Services: Process and Rules**
 - ï **Entity** beans to maintain **model data**

Assignment of Functionality to Tiers or Vertical Layering



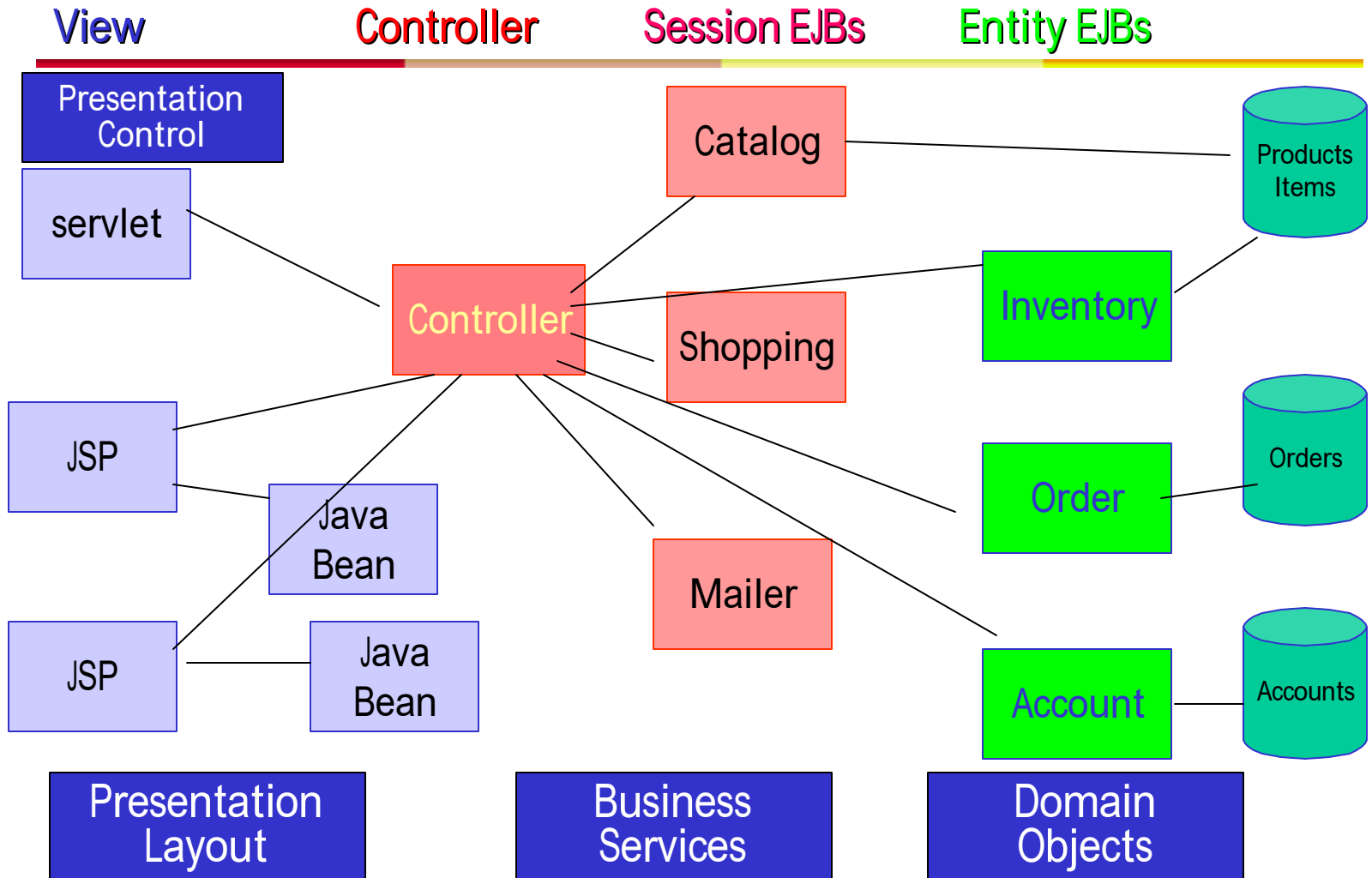
What are Domain Objects?

- Domain Objects **model the real world entities, nouns** in business problem
- Also known as **Data Model, business object model**
- Example
 - User **Account**,
 - **Order**,
 - **Product**,
 - **Inventory**

What are Business Services?

- Business Services Model the **verbs** in application
- Business processes are the **services** in the **use cases**
- Example
 - **Browse** Catalog
 - **Add** item to Cart
 - **Purchase** items

Sample Application Architecture **Vertical Layering**



Summary

- We went over:
 - Advantages of horizontal layers
 - Advantages of Vertical Tiers
 - Affect of Layers and Tiers on Capabilities.
 - the process of architecting the sample application
 - the Model View Controller Design Pattern